

# Design of a Planar First-Order Loudspeaker Array for Global Active Noise Control

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**Abstract**—This paper proposes a method to design a planar first-order loudspeaker array structure for global active noise control. Compared with the traditional spherical loudspeaker array, the planar array provides a practical design with flexible source locations. The planar array is capable of achieving global noise control, provided that the loudspeakers have general variable first-order responses in elevation. On  $x-y$  plane, we use spherical harmonics to analyze the required first-order loudspeakers consisting of monopole and tangential dipole components. By exploiting the properties of the associated Legendre functions and its derivative, we can divide the primary soundfield into even harmonics controlled by the monopole component, and odd harmonics controlled by the dipole component. Through the appropriate choice of radii of circles, we avoid the ill-conditioning problem of matrix inversion and derive a robust solution for loudspeaker weights to suppress the primary noise field. Besides, we use the closely-located monopole pairs, instead of the ideal general first-order loudspeakers, to design an alternative planar array for practical implementation. As an illustration, we use several simulation examples to validate the performance of the two proposed planar loudspeaker arrays.

**Index Terms**—Active noise control, spherical harmonics, planar loudspeaker array, general first-order sources.

## I. INTRODUCTION

ACOUSTIC noise problems are becoming increasingly ubiquitous in industry and daily life. The methods to acoustic noise control can be broadly grouped into two categories: passive methods and active noise control (ANC). The former applies the acoustic insulation materials to produce a modest attenuation over a broadband frequency range [1], [2]. However, since the low-frequency noise possesses strong penetrating capability, the acoustic insulation is relatively ineffective and costly. By contrast, ANC attempts to introduce the anti-noise wave through an appropriate array of secondary sources to reduce the primary (unwanted) noise levels, based on the principle of superposition [3]–[5]. The ANC systems have been used in many different applications, where the noise fields are

dominated by the low frequencies, such as engine noise in automobile and mechanical vibration in manufacturing.

The earlier researches with respect to ANC utilized a single loudspeaker as the secondary source to generate anti-noise of equal amplitude and opposite phase at some particular control points. Through cancelling the acoustic pressure at one control point, a monopole secondary source only can create a  $-10$  dB local quiet zone with the extent of a tenth of a wavelength [6]. A larger quiet zone can be achieved when two closely-located loudspeakers are used to cancel the pressure and particle velocity [7]. However, for attenuating global noise level, it is physically impossible that the loudspeaker requires to be co-located with the primary noise source.

With the use of a multipole source, it becomes possible for multichannel ANC systems to deal with noise fields over a large space. It is because of high complex directivity of the multipole source. The first known ANC system with deployment of a secondary multipole source was originally suggested by Kempton [8], where global noise cancellation was able to achieve outside the region of primary noises using Taylor series expansion. Thereafter, Bolton *et al.* [9] derived a closed-form solution for secondary multipole source strengths by minimizing the total soundfield power and further improved noise attenuation performance. On the basis of the above works, a corresponding experimental investigation [10] showed that a secondary multipole could provide far greater cancellation compared to a secondary monopole source at the same position. However, the design of multipole sources is difficult to achieve in practice.

A spherical loudspeaker array can be viewed as a straightforward realization for the theoretical model of a multipole source [11], [12]. As the most common and typical array structure, spherical array configuration has been investigated and employed as a multichannel spatially-compact source, by driving each loudspeaker unit independently [13]–[15]. Owing to its ability of producing sound radiation pattern with high complexity in arbitrary directions, it becomes useful for ANC system. The previous works [16], [17] compared the noise attenuation performances using an open-sphere spherical array of monopoles and a multipole source respectively, and found that the results of the spherical array of monopoles are compared with that of the multipole source. However, the open-sphere configuration cannot produce corresponding anti-waves at certain frequencies due to the zeros of spherical Bessel function [18], [19].

Another usual spherical array configuration is the rigid-sphere with loudspeakers arranged around a rigid sphere. With this

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configuration, Azarpeyvand [20] made use of spherical harmonics to represent the primary soundfield, so that the minimization efficiency of such noise-control systems was obviously improved, especially for a dipolar primary source. Rafaely [21] also took advantage of spherical harmonics to present a theoretical analysis of a spherical loudspeaker array, both for global and local ANC systems. Still in [21], it has been found that the natural quiet zones have a shell-shape for local ANC, much larger than the conventional limit of a tenth of a wavelength achieved using monopole sources. Based on the Rafaely's work, a further study and investigation was shown in [22], where a combination of two compact spherical sources was employed to enlarge the quiet zone and enhance the numerical robustness of such ANC systems. However, achieving the global ANC is required to place the primary sources very near the rigid-sphere secondary sources, which severely restricts the spatial distributions of the primary sources.

In recent years, cylindrical/spherical harmonics have been widely used for adaptive wave-domain ANC over 2D/3D space [23]–[27], where a series of multichannel ANC systems have been developed to achieve sound control over a large region. During the adaptation process, the orthogonal properties of circular/spherical harmonics were used to decouple the loudspeaker signals and further reduced the computational complexity. In order to achieve the sound control over 3D space, the spherical loudspeaker array is a natural choice as the secondary source to generate anti-noise because its geometrical structure is consistent with spherical harmonics. However, the open-sphere and the rigid-sphere spherical array both have some inherent drawbacks as mentioned above.

In contrast to the above spherical arrays, a two-dimensional planar array configuration [28] have been proposed for the purpose of 3D soundfield analysis. In this configuration, the first-order microphone is used innovatively to measure the soundfield components that are normally “invisible” for conventional planar array. In this paper, we take the original inspiration from [28] to develop an open planar first-order loudspeaker array structure for global ANC. Unlike the rigid-sphere, the open planar array structure with loudspeakers on  $x$ - $y$  plane ensures that the primary noises are allowed to be distributed widely inside the volume of a sphere. Besides, it arranges the loudspeakers on the multiple co-centered circular array layout, which overcomes the Bessel zeros problem in the open-sphere. Since the monopole sources on  $x$ - $y$  plane cannot control the odd harmonics of primary soundfields, a general first-order loudspeaker consisting of a monopole and a tangential dipole are derived and used in the proposed planar array, thus allowing the odd harmonics to be controlled by the dipole component. We demonstrate that the proposed planar loudspeaker array provides the same functionality as spherical loudspeaker arrays designed for global ANC.

The remainder of this paper is organized as follows: Section II presents spherical harmonic description of primary sources and general first-order loudspeakers. Section III designs the planar first-order loudspeaker array and employs it to achieve the global noise control. An alternative planar array of monopole pairs is proposed in Section IV, which avoids the need for dipole

loudspeakers. The several simulation examples are given in Section V to validate the noise reduction performance of the two proposed planar loudspeaker arrays, while the conclusion is drawn in Section VI.

## II. SPHERICAL HARMONIC DESCRIPTION OF 3D SOUNDFIELDS

In this Section, we first introduce spherical harmonic analysis to describe the exterior soundfields produced by the primary sources. We also derive the spherical harmonic form of the general first-order sources and further investigate the case of placing the sources on the  $x$ - $y$  plane.

### A. The Primary Sources

In a spherical coordinate, the exterior soundfields at arbitrary control point  $\mathbf{x} = (r, \theta, \phi)$  due to the primary noises has the general form [11],

$$S_{\text{pri}}(\mathbf{x}, k) = \sum_{n=0}^{\infty} \sum_{m=-n}^n \alpha_{nm}^{(\text{pri})}(k) h_n(kr) Y_{nm}(\theta, \phi), \quad (1)$$

where  $\alpha_{nm}^{(\text{pri})}(k)$  are a set of primary soundfield coefficients,  $k = 2\pi f/c$  is the wave number (with  $f$  the frequency and  $c$  the speed of wave propagation), and  $h_n(\cdot)$  is the first kind spherical Hankel function of order  $n$ . The spherical harmonic  $Y_{nm}(\cdot)$  is defined as

$$Y_{nm}(\theta, \phi) = \mathcal{P}_{nm}(\cos \theta) E_m(\phi), \quad (2)$$

where  $\mathcal{P}_{nm}(\cdot)$  is the normalized associated Legendre function and  $E_m(\phi) = e^{im\phi}/\sqrt{2\pi}$  is the normalized complex exponential function.

The primary soundfield coefficients  $\alpha_{nm}^{(\text{pri})}(k)$  characterize the exterior soundfields caused by the primary sources. It is meaningless to represent the exterior soundfields using an infinite number of coefficients. When all primary sources are located inside a spherical region  $\Omega_s$  of radius  $R_s$ , these coefficients can be truncated at  $N = \lceil keR_s/2 \rceil$  due to the natural truncation property [29], where  $\lceil \cdot \rceil$  denotes rounding up to the nearest integer. Hence, a total number of  $(N+1)^2$  coefficients are sufficient to represent the exterior soundfields.

These coefficients may be readily obtained from samples of the soundfields measured on a sphere outside the source region. Mathematically, the coefficients can be found by applying the orthogonality property of spherical harmonics to (1) as

$$\alpha_{nm}^{(\text{pri})}(k) = \frac{1}{h_n(kR_s)} \int_0^{2\pi} \int_0^\pi S_{\text{pri}}(\mathbf{x}, k) Y_{nm}^*(\theta, \phi) \sin \theta d\theta d\phi, \quad (3)$$

where  $S(\mathbf{x}, k)$  is measured on the sphere of radius  $R_s$ , and  $(\cdot)^*$  denotes the complex conjugate operation. Readers are further referred to [30] for more details on sampling strategy and measurement procedure.

### B. The General First-order Loudspeakers

The spherical harmonic description of the exterior soundfields due to an ideal monopole with position  $\mathbf{x}_s = (r_s, \theta_s, \varphi_s)$ , with

$r_s < R_s$ , is given by [31]

$$S_0(\mathbf{x}, k) = \frac{e^{ik\|\mathbf{x}-\mathbf{x}_s\|}}{4\pi\|\mathbf{x}-\mathbf{x}_s\|} \\ = \sum_{n=0}^{\infty} \sum_{m=-n}^n ikj_n(kr_s)Y_{nm}^*(\theta_s, \varphi_s)h_n(kr)Y_{nm}(\theta, \phi), \quad (4)$$

where  $j_n(\cdot)$  is the first kind of spherical Bessel function of order  $n$ . Because the monopole is restricted to the source region  $\Omega_s$ , the expansion in (4) should be limited to  $n \in [-N, N]$  [similar to the case of (1)].

A tangential dipole at position  $\mathbf{x}_s$  and oriented along  $\vec{\theta}_s$  (the direction of increasing  $\theta_s$ ) produces the soundfield, as follows, [32]

$$S_{\theta_s}(\mathbf{x}, k) = \frac{\partial S_0(\mathbf{x}, k)}{\partial l_{\theta_s}} \\ = -ik \frac{e^{ik\|\mathbf{x}-\mathbf{x}_s\|}}{4\pi\|\mathbf{x}-\mathbf{x}_s\|} \left[ 1 + \frac{i}{k\|\mathbf{x}-\mathbf{x}_s\|} \right] \cos \psi, \quad (5)$$

where  $\psi$  is the angle between  $\mathbf{x} - \mathbf{x}_s$  and  $\vec{\theta}_s$ , and  $\partial l_{\theta_s}$  is an infinitesimal length in direction  $\vec{\theta}_s$ .

In order to produce the flat response that is independent of frequency, the dipole can be equalized by dividing by  $ik$  [33]. The spherical harmonic expansion of the  $1/(ik)$ -equalized tangential dipole is obtained from the derivative of (4) with respect to  $\theta_s$  and the fact that  $\partial S_0/\partial l_{\theta_s} = \partial S_0/r_s \partial \theta_s$ ,

$$S_{\theta_s}(\mathbf{x}, k) = \frac{1}{ikr_s} \frac{\partial S_0(\mathbf{x}, k)}{\partial \theta_s} \\ = -\frac{\sin \theta_s}{r_s} \sum_{n=0}^N \sum_{m=-n}^n j_n(kr_s) \mathcal{P}'_{nm}(\cos \theta_s) \\ \times E_{-m}(\varphi_s) h_n(kr) Y_{nm}(\theta, \phi), \quad (6)$$

where  $\mathcal{P}'_{nm}(\cdot) = d\mathcal{P}_{nm}(\cdot)/d(\cdot)$  is the first-order derivative of  $\mathcal{P}_{nm}(\cdot)$ .

The general first-order source can be considered as a weighted sum of a monopole component and an equalized tangential dipole component, yielding,

$$S_{\delta, \gamma}(\mathbf{x}, k) = \delta S_0(\mathbf{x}, k) + \gamma S_{\theta_s}(\mathbf{x}, k), \quad (7)$$

where  $\delta \in [0, 1]$  and  $\gamma \in [0, 1]$  are the monopole and dipole weighting factors, respectively. By substituting (4) and (6) into (7),  $P_{\delta, \gamma}(\mathbf{x}, k)$  has the following spherical harmonic form,

$$S_{\delta, \gamma}(\mathbf{x}, k) \\ = \sum_{n=0}^N \left[ \sum_{m=-n}^n \delta ik \mathcal{P}_{nm}(\cos \theta_s) - \frac{\gamma \sin \theta_s}{r_s} \mathcal{P}'_{nm}(\cos \theta_s) \right] \\ \times E_{-m}(\varphi_s) j_n(kr_s) h_n(kr) Y_{nm}(\theta, \phi). \quad (8)$$

Through individually adjusting the values of  $\delta$  and  $\gamma$ , the general first-order source has variable first-order polar responses in the direction  $\vec{\theta}_s$ . By setting  $\gamma = 1 - \delta$ , it produces a first-order fixed-directivity polar responses, which can be thought of as a special case of the general first-order source.

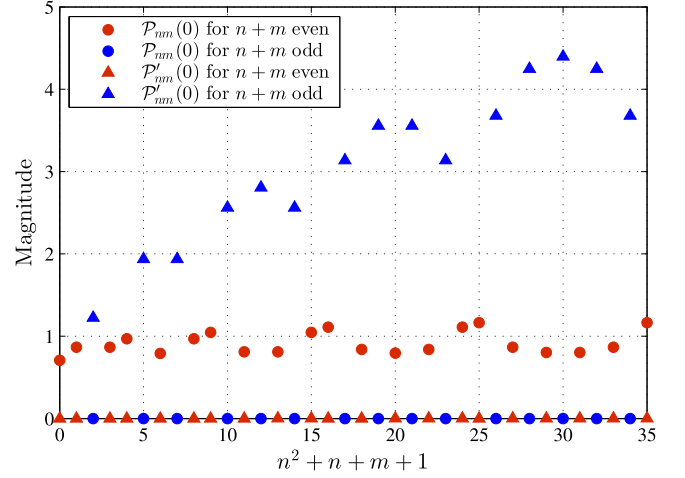


Fig. 1. Magnitude of the normalized associated Legendre function  $\mathcal{P}_{nm}(0)$  (circular markers) and its first-order derivative  $\mathcal{P}'_{nm}(0)$  (triangular markers), for orders up to  $N = 5$ . Red and blue markers indicate even terms ( $n + m = \text{even}$ ) and odd terms ( $n + m = \text{odd}$ ), respectively.

### C. The General First-order Loudspeakers on the $x$ - $y$ Plane

In what follows, we investigate the case of placing the loudspeakers on the  $x$ - $y$  plane. In this case, the co-latitude angle are equal to  $\pi/2$ . Thus, the produced soundfield of the general first-order sources at  $\theta_s = \pi/2$  (i.e.,  $\cos \theta_s = 0$ ) is

$$S_{\delta, \gamma}(\mathbf{x}, k) = \sum_{n=0}^N \sum_{m=-n}^n \left[ \delta ik \mathcal{P}_{nm}(0) - \frac{\gamma}{r_s} \mathcal{P}'_{nm}(0) \right] \\ \times E_{-m}(\varphi_s) j_n(kr_s) h_n(kr) Y_{nm}(\theta, \phi). \quad (9)$$

The function  $\mathcal{P}_{nm}(0)$  and its first-order derivative  $\mathcal{P}'_{nm}(0)$  have the following analytical expression [34],

$$\mathcal{P}_{nm}(0) = 2^m \Lambda_{nm} \cos \left[ \frac{1}{2} \pi(n+m) \right] \frac{\Gamma(\frac{1}{2}n + \frac{1}{2}m + \frac{1}{2})}{\Gamma(\frac{1}{2}n - \frac{1}{2}m + 1)}, \quad (10)$$

and

$$\mathcal{P}'_{nm}(0) = 2^{m+1} \Lambda_{nm} \sin \left[ \frac{1}{2} \pi(n+m) \right] \frac{\Gamma(\frac{1}{2}n + \frac{1}{2}m + 1)}{\Gamma(\frac{1}{2}n - \frac{1}{2}m + \frac{1}{2})}, \quad (11)$$

with

$$\Lambda_{nm} = (-1)^m \sqrt{\frac{2n+1}{2\pi}} \sqrt{\frac{(n-m)!}{(n+m)!}}, \quad (12)$$

where  $\Gamma(\cdot)$  denotes the Gamma function. Fig. 1 illustrates the normalized associated Legendre function  $\mathcal{P}_{nm}(0)$  and its first-order derivative  $\mathcal{P}'_{nm}(0)$  for orders up to  $N = 5$ . Since the cosine term is included in (10), we have  $\mathcal{P}_{nm}(0) = 0$  if  $n + m$  is odd (blue circular markers). Similarly, since the sine term is included in (11), we have  $\mathcal{P}'_{nm}(0) = 0$  if  $n + m$  is even (red triangular markers). The underlying property indicates that the monopole component in (9) has zero contribution to the odd harmonics of the produced soundfield, and the dipole component in (9) has zero contribution to the even harmonics of the produced

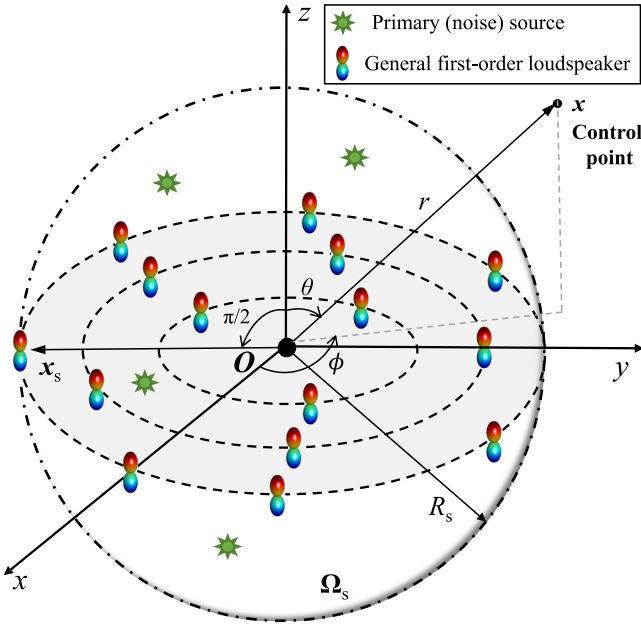


Fig. 2. Example of planar first-order loudspeaker array structure for global active noise control.

soundfield. In the following, we use the underlying property to design a planar first-order loudspeaker array.

### III. A PLANAR FIRST-ORDER LOUDSPEAKER ARRAY CONFIGURATION

For achieving the global noise reduction, the design of a compact planar first-order loudspeaker array, shown in Fig. 2, is proposed to produce the anti-noise signals. Let the source region  $\Omega_s$  to be a sphere of radius  $R_s$  centered at the origin  $O$ , where both all primary sources and the first-order sources are located. The first-order sources are placed on the multiple co-centered circular structure on the  $x$ - $y$  plane, thus forming a compact planar loudspeaker array. Using the proposed planar array, our aim is to reduce the global noise levels in the exterior of  $\Omega_s$ , which is the control region of our interest.

#### A. Soundfields Due to the First-order Loudspeaker Array

We now consider a set of  $Q$  concentric circles of planar continuous loudspeaker located at  $(r_q, \pi/2, \varphi)$ ,  $q = 1, 2, \dots, Q$ . Let the  $\delta_q(\varphi, k)$  and  $\gamma_q(\varphi, k)$  be the aperture functions of monopole and tangential dipole components, respectively, as a periodic function of azimuth angle  $\varphi$ . Thus, the aperture functions  $\delta_q(\varphi, k)$  and  $\gamma_q(\varphi, k)$  are expressed by a Fourier series as

$$\delta_q(\varphi, k) = \sum_{m=-\infty}^{\infty} \beta_{m,q}^{(e)}(k) E_m(\varphi), \quad (13)$$

and

$$\gamma_q(\varphi, k) = \sum_{m=-\infty}^{\infty} \beta_{m,q}^{(o)}(k) E_m(\varphi), \quad (14)$$

where  $\beta_{m,q}^{(e)}(k)$  and  $\beta_{m,q}^{(o)}(k)$  are the Fourier coefficients of the aperture functions  $\delta_q(\varphi, k)$  and  $\gamma_q(\varphi, k)$ , respectively.

Thus, the corresponding soundfield at a control point  $\mathbf{x} = (r, \theta, \phi)$  due to the planar array are given by integrating (7) with respect to  $\varphi$  over  $[0, 2\pi)$  and summing the soundfield contributions from all circles,

$$S_{\text{sec}}(\mathbf{x}, k) = \sum_{q=1}^Q \left\{ \int_0^{2\pi} \left[ \delta_q(\varphi, k) S_0(\mathbf{x}, k) + \gamma_q(\varphi, k) S_{\theta_s}(\mathbf{x}, k) \right] d\varphi \right\}. \quad (15)$$

Then, by substituting (9), (13) and (14) into (15) and employing the orthogonality property of the exponential functions, the soundfield produced by the planar array can be represented in the form of spherical harmonics as,

$$S_{\text{sec}}(\mathbf{x}, k) = \sum_{q=1}^Q \sum_{n=0}^N \sum_{m=-n}^n \left[ \beta_{m,q}^{(e)}(k) i k \mathcal{P}_{nm}(0) - \frac{\beta_{m,q}^{(o)}(k)}{r_q} \mathcal{P}'_{nm}(0) \right] j_n(kr_q) h_n(kr) Y_{nm}(\theta, \phi). \quad (16)$$

Since the underlying property of the associated Legendre function and its derivative in Sec II-C, we can remove the odd harmonics in monopole component and the even harmonics in dipole component and rewrite (16) as

$$\begin{aligned} S_{\text{sec}}(\mathbf{x}, k) &= \sum_{\substack{n=0 \\ n+m=\text{even}}}^N \sum_{m=-n}^n \sum_{q=1}^Q \beta_{m,q}^{(e)}(k) i k j_n(kr_q) \mathcal{P}_{nm}(0) h_n(kr) Y_{nm}(\theta, \phi) \\ &\quad - \sum_{\substack{n=0 \\ n+m=\text{odd}}}^N \sum_{m=-n}^n \sum_{q=1}^Q \frac{\beta_{m,q}^{(o)}(k)}{r_q} j_n(kr_q) \mathcal{P}'_{nm}(0) h_n(kr) Y_{nm}(\theta, \phi). \end{aligned} \quad (17)$$

Note that the even and odd harmonics of the produced soundfields are determined respectively by the aperture function coefficients  $\beta_{m,q}^{(e)}(k)$  and  $\beta_{m,q}^{(o)}(k)$ . In SubSection III-B and III-C, We derive the solutions for these aperture function coefficients  $\beta_{m,q}^{(e)}(k)$  and  $\beta_{m,q}^{(o)}(k)$  to suppress the even and odd harmonics of the primary soundfields.

#### B. Active Control of Even Harmonics

We design the aperture function coefficients  $\beta_{m,q}^{(e)}(k)$  so that the even harmonics of the primary soundfields are suppressed outside the source region. By considering the even harmonic terms of (1) and (17) in terms of soundfield coefficients, the active control of even harmonics can be achieved by satisfying the condition

$$\alpha_{nm}^{(\text{pri})}(k) + \sum_{q=1}^Q \beta_{m,q}^{(e)}(k) i k j_n(kr_q) \mathcal{P}_{nm}(0) = 0, \quad (18)$$



for  $m = -N, \dots, N$  and  $n = |m|, |m| + 2, \dots, N_e$ . And  $N_e$  is equal to  $N$  if  $m + N$  is even, otherwise it is equal to  $N - 1$ . Note that the even harmonics of the primary soundfields may be counteracted by controlling the Fourier coefficients  $\beta_{m,q}^{(e)}$ .

For a particular value of  $m$ , (18) can be written in matrix form, having the following simultaneous equation as

$$\alpha_m^{(e)} = \mathbf{T}_m \beta_m^{(e)}, \text{ for } m = -N, \dots, N, \quad (19)$$

$$\text{where } \alpha_m^{(e)} = - \left[ \alpha_{|m|}^{(\text{pri})}(k), \alpha_{|m|+2}^{(\text{pri})}(k), \dots, \alpha_{N_e m}^{(\text{pri})}(k) \right]^T,$$

$$\mathbf{T}_m = ik$$

$$\times \begin{bmatrix} j_{|m|}(kr_1) \mathcal{P}_{|m|m}(0) & \cdots & j_{|m|}(kr_Q) \mathcal{P}_{|m|m}(0) \\ j_{|m|+2}(kr_1) \mathcal{P}_{|m|+2m}(0) & \ddots & j_{|m|+2}(kr_Q) \mathcal{P}_{|m|+2m}(0) \\ \vdots & \ddots & \vdots \\ j_{N_e}(kr_1) \mathcal{P}_{N_e m}(0) & \cdots & j_{N_e}(kr_Q) \mathcal{P}_{N_e m}(0) \end{bmatrix} \quad (20)$$

and  $\beta_m^{(e)} = \left[ \beta_{m,1}^{(e)}(k), \beta_{m,2}^{(e)}(k), \dots, \beta_{m,Q}^{(e)}(k) \right]^T$ . The vector of the Fourier coefficients  $\beta_m^{(e)}$  could be estimated using the least-squares method provided that  $\mathbf{T}_m$  is nonsingular, as

$$\beta_m^{(e)} = \mathbf{T}_m^\dagger \alpha_m^{(e)}, \quad (21)$$

where  $(\cdot)^\dagger$  denotes the Moore-Penrose inverse [35]. Observe that the structure of the matrix  $\mathbf{T}_m$  is only related to the radii of circles. If we choose the radii arbitrarily, the resulting solutions may be non-robust and result in ill-conditioning problem. Although Tikhonov regularization is the most commonly method for obtaining the robust solutions for  $\beta_{m,q}^{(e)}(k)$  [36], a more reasonable method is to appropriately choose the radii of circles rather than only relying on the ability of Tikhonov regularization to avoid singularities of  $\mathbf{T}_m$ . The details about the choice of circles will be investigated in SubSection III-D, which guarantees that the even harmonics of the primary soundfields are controlled effectively by  $\beta_{m,q}^{(e)}(k)$ .

### C. Active Control of Odd Harmonics

Active control of even harmonics can achieve the noise reduction on the  $x$ - $y$  plane. However, it cannot totally reduce the out-of-plane noise level due to the contributions from the odd harmonics of the primary soundfields. We design the aperture function coefficients  $\beta_{m,q}^{(o)}(k)$  so that the odd harmonics are suppressed outside the source region. By considering the odd harmonic terms of (1) and (17) in terms of soundfield coefficients, the active control of odd harmonics can be achieved by satisfying the condition

$$\alpha_{nm}^{(\text{pri})}(k) - \sum_{q=1}^Q \frac{\beta_{m,q}^{(o)}(k)}{r_q} j_n(kr_q) \mathcal{P}'_{nm}(0) = 0, \quad (22)$$

for  $m = -N, \dots, N$  and  $n = |m| + 1, |m| + 3, \dots, N_o$ . And  $N_o$  is equal to  $N - 1$  if  $m + N$  is even, otherwise it is equal to  $N$ . Note that the odd harmonics of the primary soundfields may be counteracted by controlling the Fourier coefficients  $\beta_{m,q}^{(o)}(k)$ .

For a particular value of  $m$ , (22) can be written in matrix form, having the following simultaneous equation as

$$\alpha_m^{(o)} = \mathbf{J}_m \mathbf{D} \beta_m^{(o)}, \text{ for } m = -N, \dots, N, \quad (23)$$

$$\text{where } \alpha_m^{(o)} = \left[ \alpha_{|m|+1}^{(\text{pri})}(k), \alpha_{|m|+3}^{(\text{pri})}(k), \dots, \alpha_{N_o m}^{(\text{pri})}(k) \right]^T,$$

$$\mathbf{D} = \text{diag} [1/r_1, 1/r_2, 1/r_3, \dots, 1/r_Q], \quad (24)$$

$$\mathbf{J}_m =$$

$$\begin{bmatrix} j_{|m|+1}(kr_1) \mathcal{P}'_{|m|+1m}(0) & \cdots & j_{|m|+1}(kr_Q) \mathcal{P}'_{|m|+1m}(0) \\ j_{|m|+3}(kr_1) \mathcal{P}'_{|m|+3m}(0) & \ddots & j_{|m|+3}(kr_Q) \mathcal{P}'_{|m|+3m}(0) \\ \vdots & \ddots & \vdots \\ j_{N_o}(kr_1) \mathcal{P}'_{N_o m}(0) & \cdots & j_{N_o}(kr_Q) \mathcal{P}'_{N_o m}(0) \end{bmatrix} \quad (25)$$

and  $\beta_m^{(o)} = \left[ \beta_{m,1}^{(o)}(k), \beta_{m,2}^{(o)}(k), \dots, \beta_{m,Q}^{(o)}(k) \right]^T$ . The vector of the Fourier coefficients  $\beta_m^{(o)}$  could be estimated using the least-squares method provided that both  $\mathbf{D}$  and  $\mathbf{J}_m$  are nonsingular, as

$$\beta_m^{(o)} = \mathbf{D}^{-1} \mathbf{J}_m^\dagger \alpha_m^{(o)}. \quad (26)$$

The diagonal matrix  $\mathbf{D}$  is nonsingular because the diagonal elements are all non-zero. As with  $\mathbf{T}_m$  in (20), the structure of the matrix  $\mathbf{J}_m$  is seen to depend on the radii of circles. In the next Subsection, we will ensure that the matrices  $\mathbf{T}_m$  and  $\mathbf{J}_m$  are both well-conditioned by appropriately choosing the radii of circles.

### D. Array placement

1) *The choice of radii*: The choice of radii impacts the validity of Moore-Penrose inverse for both  $\mathbf{T}_m$  and  $\mathbf{J}_m$ . The well-conditioned matrices require that the radii of circles should have a good diversity, similar to the previous works [28], [37]. Specifically, we choose the radii of circles uniformly within the target frequency band  $k \in [k_l, k_u]$  as

$$r_q = \frac{q}{Q} R_s, \text{ for } q = 1, 2, 3, \dots, Q, \quad (27)$$

To guarantee that the simultaneous (19) and (23) can be solved exactly, we need place at least  $Q = \lceil (N_{\max} + 1)/2 \rceil$  circles on the  $x$ - $y$  plane, where  $N_{\max} = \lceil k_u R_s / 2 \rceil$  is the maximum truncation order. Due to the high-pass nature of the spherical Bessel functions, the higher order Bessel functions go to zero on smaller radii circles. Therefore, this choice can make the structures of  $\mathbf{T}_m$  and  $\mathbf{J}_m$  become close to upper triangular matrices, which ensure the validity of Moore-Penrose inverse. With the choice of radii, the order of the exterior soundfield on the  $q$ th circle is limited to  $N_q = \lceil k r_q / 2 \rceil$ . In order to further improve the stability of the solutions, the extra circles could be recommended to be included in the proposed planar array.

2) *Sampling of circles*: For a practical loudspeaker array, the continuous aperture functions cannot directly drive the loudspeaker signals. They need to be discretized to derive the loudspeaker weights. According to the Shannon sampling theorem,

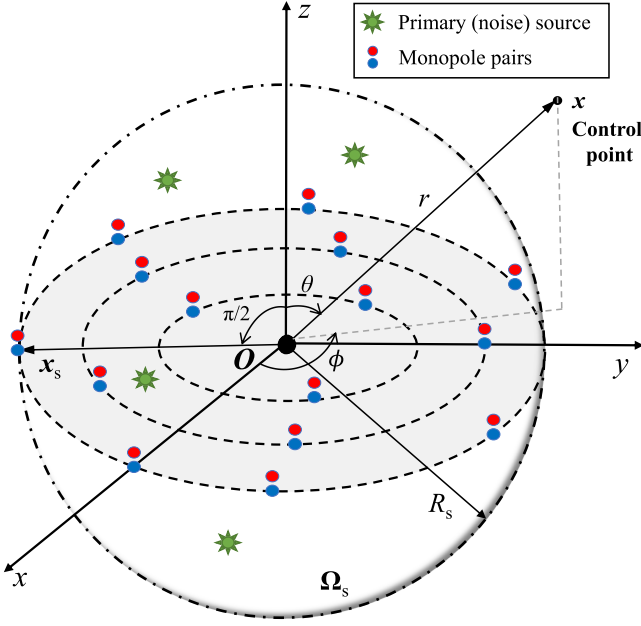


Fig. 3. Example of an alternative planar array structure consisting of monopole pairs for global active noise control. The lower monopoles are marked by blue solid circles, and the upper monopoles are marked by red solid circles.

we should place  $L_q \geq 2N_q + 1$  loudspeakers equiangularly on the  $q$ th circle, so as to exactly reproduce the aperture functions  $\delta_q(\varphi, k)$  and  $\gamma_q(\varphi, k)$  [38]. Thus, the monopole and dipole weights of the  $l$ th loudspeaker on the  $q$ th circle are given by

$$\delta_q(\varphi_{ql}, k) = \sum_{m=-N_q}^{N_q} \beta_{m,q}^{(o)}(k) E_m(\varphi_{ql}) \Delta_q, \quad (28)$$

$$\gamma_q(\varphi_{ql}, k) = \sum_{m=-N_q}^{N_q} \beta_{m,q}^{(e)}(k) E_m(\varphi_{ql}) \Delta_q, \quad (29)$$

where  $\Delta_q = 2\pi/L_q$  is the angular space of the loudspeakers and  $\varphi_{ql} = l\Delta_q$  results from uniform sampling. Nonuniform sampling could also be recommended as long as the maximum angular space is not larger than  $\Delta_q$ .

#### IV. AN ALTERNATIVE PLANAR ARRAY CONSISTING OF MONOPOLE PAIRS

An alternative array configuration, shown in Fig. 3, is to use the monopole pairs separated by an infinitesimal distance oriented along the negative  $z$  axis, instead of the general first-order sources. In order to produce the same exterior soundfields as (9), there are two different purposes for each monopole pair: the two monopole sources are used together to form a dipole component for suppressing the odd harmonics; besides, the lower monopole source is used to produce the monopole component for suppressing the even harmonics. In this way, the general first-order sources can be replaced with the closely-located monopole sources, which are widely used in practical loudspeaker arrays.

Assuming the distance  $\Delta z$  is vanishingly small ( $\Delta z \leq 1/k_u$ ), the upper and lower monopoles are differentiated to approximate

the dipole oriented along the negative  $z$  axis [11]

$$-\frac{\partial S_0(\mathbf{x}, k)}{\partial z} \approx \frac{S_0^{(\text{lower})}(\mathbf{x}, k) - S_0^{(\text{upper})}(\mathbf{x}, k)}{\Delta z}, \quad (30)$$

where  $S_0^{(\text{lower})}$  denotes the soundfields from the lower monopole, and  $S_0^{(\text{upper})}$  denotes the soundfields from the upper monopole. Because the monopole pairs are located on  $x$ - $y$  plane, the negative  $z$  axis has the same direction with  $\theta_s$ , so as to  $S_{\theta_s}(\mathbf{x}, k) = -\partial S_0(\mathbf{x}, k)/(\partial z)$ .

Following the array placement in SubSection III-D, the weights of the  $l$ th monopole pairs on the  $q$ th circle are directly linked to that of the general first-order sources

$$w_q^{(\text{lower})}(\varphi_{ql}, k) = \delta_q(\varphi_{ql}, k) + \frac{\gamma_q(\varphi_{ql}, k)}{ik\Delta z}, \quad (31)$$

$$w_q^{(\text{upper})}(\varphi_{ql}, k) = -\frac{\gamma_q(\varphi_{ql}, k)}{ik\Delta z}, \quad (32)$$

where  $w_q^{(\text{lower})}(\varphi_{ql}, k)$  denotes the weight of the lower monopoles, and  $w_q^{(\text{upper})}(\varphi_{ql}, k)$  denotes the weight of the upper monopoles. Note that the lower monopoles not only contribute to control the even harmonics, but also contribute to control the odd harmonics together with the upper monopoles.

#### V. SIMULATION EXAMPLES

In this section, we present several examples of the two proposed planar loudspeaker array for mono-frequency and broadband situations. Two types of primary sources are considered: 1) a simple monopole source and 2) a complex source composed of 10 monopoles arranged uniformly inside the source region. In the mono-frequency examples, we use a specific fifth-order planar array to validate the reduction performance for a monopole primary source. We also consider the broadband performance for complex primary source within a design frequency band [20, 800] Hz.

In order to quantify the performance of the proposed planar array, we define the noise reduction level averaged over all angles [39], [40], as follows,

$$\varepsilon(r, k) = 10 \log_{10} \frac{\int_0^{2\pi} \int_0^\pi |S_{\text{pri}}(\mathbf{x}, k)|^2 \sin \theta d\theta d\phi}{\int_0^{2\pi} \int_0^\pi |S_{\text{pri}}(\mathbf{x}, k) + S_{\text{sec}}(\mathbf{x}, k)|^2 \sin \theta d\theta d\phi}. \quad (33)$$

Employing the orthogonality property of spherical harmonics and the spherical harmonic decomposition in (1) and (9), the noise reduction level can be written in terms of soundfield coefficients

$$\begin{aligned} \varepsilon(r, k) &= 10 \log_{10} \frac{\sum_{n=0}^N |h_n(kr)|^2 \sum_{m=-n}^n |\alpha_{nm}^{(\text{pri})}(k)|^2}{\sum_{n=0}^N |h_n(kr)|^2 \sum_{m=-n}^n |\alpha_{nm}^{(\text{pri})}(k) + \alpha_{nm}^{(\text{sec})}(k)|^2}, \end{aligned} \quad (34)$$

where the secondary soundfield coefficients  $\alpha_{nm}^{(\text{sec})}(k)$  for the first-order loudspeaker array are given (35), and  $\alpha_{nm}^{(\text{sec})}(k)$  for

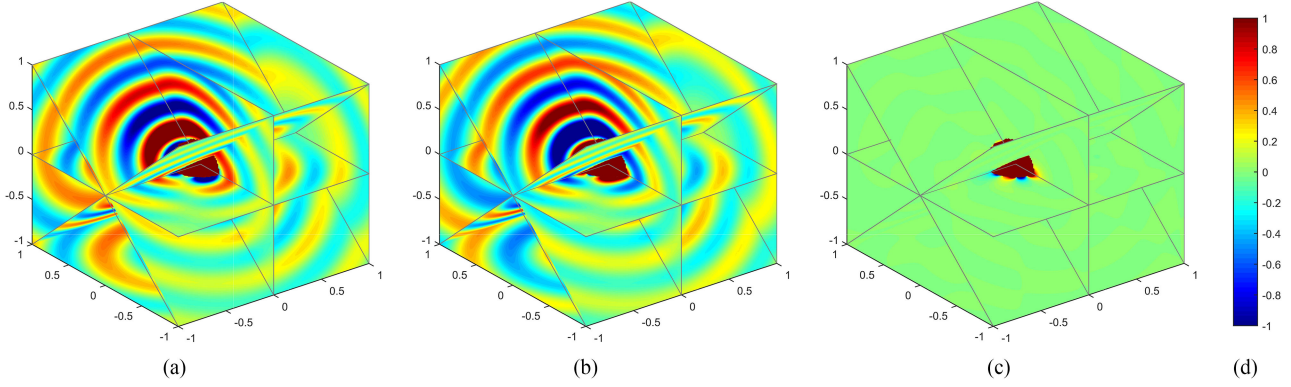


Fig. 4. A comparison of (a) primary soundfields, (b) anti-noise soundfields and (c) residual soundfields for an 800-Hz primary complex source composed of 10 monopoles arranged uniformly on the  $x$ - $y$  plane. Anti-noise and residual soundfields are only controlled by the monopole component of the proposed planar array. The dark red circular region denotes the source region where primary sources lie.

the alternative planar array in Section IV are given in (36).

$$\alpha_{nm}^{(\text{sec})}(k) = \begin{cases} \sum_{q=1}^Q \sum_{l=1}^l \delta_q(\varphi_{ql}, k) ik \mathcal{P}_{nm}(0) E_{-m}(\varphi_{ql}) j_n(kr_q), & \text{for } n+m = \text{even} \\ - \sum_{q=1}^Q \sum_{l=1}^l \frac{\gamma_q(\varphi_{ql}, k)}{r_q} \mathcal{P}'_{nm}(0) E_{-m}(\varphi_{ql}) j_n(kr_q), & \text{for } n+m = \text{odd} \end{cases} \quad (35)$$

$$\alpha_{nm}^{(\text{sec})}(k) = \sum_{q=1}^Q \sum_{l=1}^l ik E_{-m}(\varphi_{ql}) \left[ w_q^{(\text{lower})}(\varphi_{ql}, k) j_n(kr_q) \mathcal{P}_{nm}(0) + w_q^{(\text{upper})}(\varphi_{ql}, k) j_n\left(k\sqrt{r_q^2 + \Delta z^2}\right) \mathcal{P}_{nm}\left(\frac{\Delta z}{\sqrt{r_q^2 + \Delta z^2}}\right) \right] \quad (36)$$

#### A. Mono-Frequency Performance

We firstly consider a planar loudspeaker array with maximum radius  $R_s = 0.25$  m to attenuate the primary soundfield. The primary source is assumed to be a monopole source inside the source region, and the design frequency is assumed up to  $f_u = 800$  Hz. Due to the limitation of the truncation theorem, the maximum truncation order is chosen to be  $N_{\max} = \lceil k_u e R_s / 2 \rceil = 5$ . Following the array placement in III-D, we place 3 circles of loudspeakers on the  $x$ - $y$  plane at  $r_1 = 0.083$  m,  $r_2 = 0.167$  m,  $r_3 = 0.25$  m with 5, 9, and 11 loudspeakers, respectively. Thus, we use a total of 25 first-order loudspeakers to constitute a fifth-order ( $N_{\max} = 5$ ) planar array system used for global ANC. Correspondingly, the alternative planar array is built to have 25 monopole pairs, including 25 lower monopoles and 25 upper monopoles. The distance  $\Delta z$  between the lower monopoles and the upper monopoles is set as  $\Delta z = 0.1/k_u = 6.8$  mm. For a higher frequency ( $f_u > 800$  Hz), we would require a higher order arrays and/or smaller regions of the source.

In the following simulation, we first consider an 800 Hz primary complex source composed of 10 monopoles arranged uniformly on the  $x$ - $y$  plane. Fig. 4 demonstrates a comparison of (a) primary soundfields, (b) anti-noise soundfields and (c) residual soundfields. The planar array only uses the monopole component to create a quiet zone outside the source region. The reason for this is that the primary complex source on  $x$ - $y$  plane only produces the even harmonics due to the underlying property of the associated Legendre functions. However, once the primary source is far away from the  $x$ - $y$  plane, the monopole component cannot create a quiet zone out of the  $x$ - $y$  plane due to the contribution from odd harmonics produced by the primary source, as shown in Fig. 5(b). After including the dipole component, these odd harmonics can be totally eliminated using the planar array, as shown in Fig. 5(c).

To validate the array performance for primary sources with different positions, we consider an example that a monopole primary source varies from zenith angle  $\theta_{\text{pri}} = 0^\circ$  to  $\theta_{\text{pri}} = 180^\circ$ . Because of the symmetry of the planar loudspeaker array over the horizontal azimuths, we can ignore the variation about the azimuths in this example. Fig. 6 shows the corresponding noise reduction levels as a function of zenith angle  $\theta_{\text{pri}}$  for the primary source at different radii ( $r_{\text{pri}} = 0.1$  m,  $0.2$  m,  $0.25$  m). It can be observed from Fig. 6 that in general, the two proposed planar arrays always yield higher than 18 dB noise reduction level for different radii, no matter what the zenith angles varies. Additionally, we notice that the noise reduction level gradually reduces with the increase of the radius  $r_{\text{pri}}$ , both for 400 Hz and 800 Hz primary sources. This is due to the fact that the higher order harmonics are produced when the radius of the primary source becomes large. These higher order harmonics cannot be eliminated by the inner circles within small radius. For the same reason, the noise reduction performance at the frequency of 400 Hz in Fig. 6(a) provides better results than that at the frequency of 800 Hz in Fig. 6(b), where an 800 Hz primary source can produce the higher order harmonics.

Compared to the alternative alternative planar array consisting of monopole pairs, the planar array using the general first-order sources has better noise reduction performance. For the case where the primary source is far away from the  $x$ - $y$  plane, the



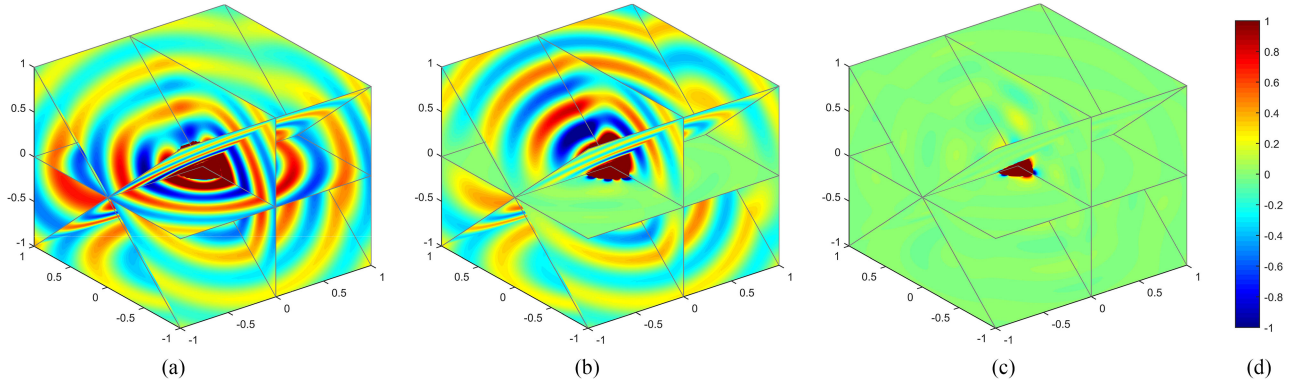


Fig. 5. A comparison of the primary and residual soundfields for an 800-Hz primary complex source composed of 10 monopoles arranged uniformly in the source region. (a) Primary soundfield, (b) residual soundfields by controlling the monopole component, and (c) residual soundfields by controlling the planar first-order loudspeaker array.

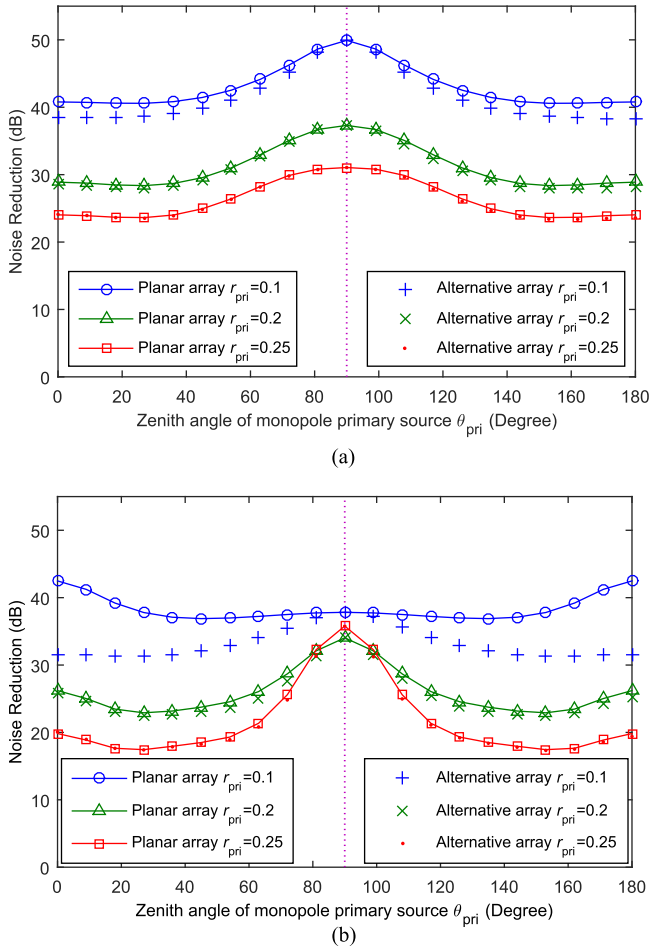


Fig. 6. Noise reduction level for monopole primary source varying from zenith angle  $\theta_{pri} = 0^\circ$  to  $\theta_{pri} = 180^\circ$ , operating at (a) 400 Hz and (b) 800 Hz.

contribution of the dipole component to the anti-noise soundfields is much larger than that of the monopole component, so that the difference between the two planar arrays is obvious. Although the general first-order sources can be replaced with the closely-located monopole pair, the implementation can only

approximate equivalent to the ideal general first-order source. For the case where the primary source is located on the  $x$ - $y$  plane, the noise reduction levels between the two planar arrays are exactly the same. This is because that all the produced exterior soundfields only come from even harmonics controlled by the monopole component. The monopole component of the first-order planar array can be replaced without distortion with the lower monopole in monopole pairs.

### B. Broadband Performance

Further considering the broadband performance, we consider the two proposed planar arrays for the source regions of different radii  $R_s$ . In order to better evaluate the array's capability, the open-sphere array is also used for comparison. The specific configurations of the planar arrays and the open-sphere array are described in Table I, where the required loudspeaker number  $L_{total}$  of the planar arrays obviously reduces compared to the open-sphere array. Since these arrays inevitably undergo position deviations during the practical placement process, we choose the maximum deviation error is 10 mm and the deviation error is uniformly distributed over  $[0, 10 \text{ mm}]$ .

Fig. 7 shows the noise reduction levels as a function of frequency obtained by using the proposed first-order planar array, the alternative planar array and the open-sphere array, respectively. As can be seen from Fig. 7, the noise reduction levels for the three arrays gradually decrease with the increase of frequency, because the higher order harmonics produce at high frequencies. Since the monopole pair only approximates the dipole component of the first-order loudspeaker, the reduction performance of the alternative planar array always slightly worse than that of the planar first-order speaker array. This is consistent with the mono-frequency performance.

On the other hand, as the designed radius  $R_s$  increases, the number of frequency bins corresponding to the spherical Bessel zeros grows quickly, as shown in Fig. 7(a)–(d). In the vicinity of these frequencies, the performance of the open-sphere array may be adversely affected and suffers from ill-conditioned numerical solutions, resulting in some very low valley values below  $-10 \text{ dB}$ . However, these valley values disappear for the



TABLE I  
THE PLANAR AND OPEN-SPHERE ARRAY CONFIGURATIONS FOR GLOBAL ACTIVE NOISE CONTROL OUTSIDE THE SPHERE REGIONS OF DIFFERENT  $R_s$

| Frequency       | $R_s$  | $N_{\max}$ | Planar array |                                        |                      | Open-sphere array |                      |
|-----------------|--------|------------|--------------|----------------------------------------|----------------------|-------------------|----------------------|
|                 |        |            | $Q$          | $r_q$ (m)                              | $L_{\text{total}}^a$ | Radius            | $L_{\text{total}}^a$ |
| [20, 800]<br>Hz | 0.25 m | 5          | 3            | 0.083, 0.167, 0.25                     | 25                   | 0.25 m            | 36                   |
|                 | 0.3 m  | 6          | 4            | 0.075, 0.150, 0.225, 0.3               | 36                   | 0.3 m             | 49                   |
|                 | 0.4 m  | 8          | 5            | 0.080, 0.160, 0.240, 0.320, 0.4        | 57                   | 0.4 m             | 81                   |
|                 | 0.5 m  | 10         | 6            | 0.083, 0.167, 0.250, 0.333, 0.417, 0.5 | 80                   | 0.5 m             | 121                  |

<sup>a</sup> The minimum required number of loudspeakers.

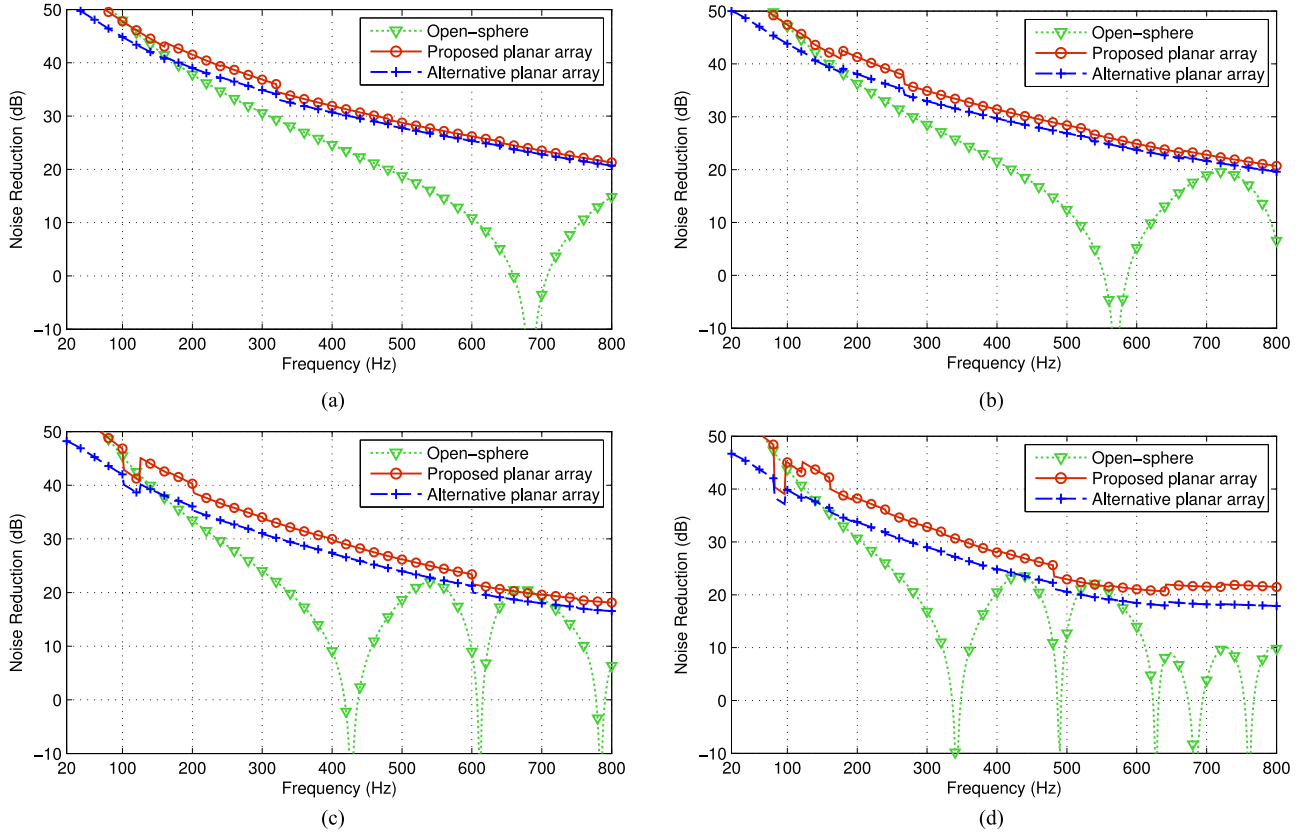


Fig. 7. Noise reduction level as a function of frequency using the proposed first-order planar array, the alternative planar array and the open-sphere array for different radii  $R_s$ : (a)  $R_s = 0.25$  m, (b)  $R_s = 0.3$  m, (c)  $R_s = 0.4$  m and (d)  $R_s = 0.5$  m.

two proposed planar arrays. The multi-radius geometry provides a potentially practical structure for avoiding the Bessel zeros problem over a wide frequency range. When the maximum frequency of interest is 800 Hz, the noise reduction level always remain higher than 18 dB both for the first-order planar loudspeaker array and the alternative planar array with different radii, indicating the effectiveness of the two proposed planar arrays.

## VI. CONCLUSION

This paper systematically developed a planar first-order loudspeaker array configuration used for global ANC. The investigation of the associated Legendre functions and its derivative

lead to the design strategy that even harmonics are controlled by monopole component and odd harmonics are controlled by the dipole component, respectively. The planar loudspeaker array not only provides another option for traditional open/rigid spherical array to suppress the primary noise field, but also has a potential ability to avoid the Bessel zeros problem. The array configuration consists of multiple co-centered circular arrays using the general first-order loudspeakers or monopoles only. We have demonstrated several practically realizable array designs to achieve global noise cancellation within the desired frequency band. Simulations results show that the two proposed planar arrays can effectively suppress the primary noises and create a quiet zone outside the source region both for mono-frequency and broadband situations.

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